

DATA ANALYTICS FINAL PROJECT

Dataset Recommendation Report & Task 1 Submission

Prepared for: Program with AI Driven Data Analytics

Submission Component: Task 1 — Problem Definition & Dataset Selection

Date: June 2026

1. Purpose of This Report

This report identifies the strongest possible dataset for the Data Analytics Final Project by evaluating candidate datasets directly against the official project guidelines: the dataset requirements (minimum 500 records, minimum 10 features, a mix of numerical and categorical data), the depth needed for Task 2 (data cleaning and pre-processing), the breadth needed for Task 3 (univariate, bivariate, and multivariate EDA with 10 or more visualizations), and the documentation and insight depth needed for Task 4 (at least five well-explained business insights).

Ten candidate datasets were researched and verified against current source listings (Kaggle and related repositories) as of June 2026. Each is ranked, profiled in full, and evaluated for ease of use, analytical richness, and viva (oral defense) friendliness. A single best recommendation is then provided, followed by a ready-to-submit Task 1 write-up formatted for direct inclusion in your project notebook.

2. Top 10 Dataset Recommendations (Ranked)

The table below ranks all ten candidates from strongest to weakest fit for this specific project's rubric. Full profiles for every dataset follow in Section 3.

Rank	Dataset	Domain	Records	Cols	Viva Score
1	Telco Customer Churn (IBM / WA_Fn-UseC_)	Telecom / Customer Analytics	7,043	21	9/10
2	IBM HR Analytics Employee Attrition & Performance	Human Resources	1,470	35	8.5/10
3	Olist Brazilian E-Commerce Public Dataset	Sales & E-commerce	~100,000	40+ (joined)	7.5/10
4	World Happiness Report (latest edition)	Social / Public Policy	~150/yr (multi-yr)	10-12	8/10
5	Airbnb Listings (NYC, Inside Airbnb / Kaggle mirror)	Retail / Real Estate Analytics	~49,000	16	8/10
6	Healthcare Patient Records (Synthetic Healthcare Dataset)	Healthcare	10,000+	15	7.5/10

Rank	Dataset	Domain	Records	Cols	Viva Score
7	Global Superstore / Sample Superstore Sales	Sales & E-commerce / Finance	~9,800	18-21	8/10
8	Students Performance in Exams	Education	1,000	8 (native)	7/10
9	Global Air Quality Index by City and Country	Environmental	10,000+ (varies)	10-12	7/10
10	IMDB Dataset of 50K Movie Reviews (Sentiment)	Social Media & Public Sentiment	50,000	2 native; 10+ engineered	6.5/10

3. Detailed Dataset Profiles

#1 — Telco Customer Churn (IBM / WA_Fn-UseC_)

Source Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Domain	Telecom / Customer Analytics
Number of Records	7,043
Number of Columns	21
Numerical Columns	Tenure, MonthlyCharges, TotalCharges, SeniorCitizen (binary flag)
Categorical Columns	Gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, Churn
Difficulty Level	Easy-Medium
Viva Friendliness Score	9/10

Why This Dataset Is Suitable

A clean, real-world, business-relevant dataset with a near-even split of numerical and categorical fields, a clear target variable (Churn), and natural groupings for every required statistical and visual technique. Exceptionally well documented with thousands of public reference notebooks for cross-checking your own logic.

Possible Preprocessing Operations

Convert TotalCharges from object to numeric (handles hidden blank strings); impute or drop the resulting nulls; encode Yes/No and SeniorCitizen flags; create a tenure-bucket derived column

(e.g., 0-12, 13-24, 25-48, 49+ months); create an Average Monthly Spend derived column; drop CustomerID before analysis.

Possible EDA Analyses

Univariate: churn rate distribution, tenure distribution, charges distribution. Bivariate: churn vs contract type, churn vs internet service, charges vs tenure. Multivariate: churn vs (contract type x payment method); correlation matrix of all numeric fields; groupby contract type and payment method for average charges; pivot table of churn rate by tenure bucket and contract type.

Possible Visualizations

Churn count bar chart, churn rate by contract type (stacked bar), tenure histogram, monthly charges boxplot by churn, correlation heatmap, scatter plot of tenure vs total charges colored by churn, pie chart of payment method share, violin plot of charges by internet service, subplot grid comparing churn drivers, line chart of cumulative charges by tenure.

Possible Insights

Which contract type has the highest churn risk; whether higher monthly charges correlate with churn; which payment method is associated with the most churn; how tenure relates to retention; which add-on services (tech support, online security) reduce churn likelihood.

#2 — IBM HR Analytics Employee Attrition & Performance

Source Link: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Domain	Human Resources
Number of Records	1,470
Number of Columns	35
Numerical Columns	Age, DailyRate, DistanceFromHome, MonthlyIncome, MonthlyRate, PercentSalaryHike, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, NumCompaniesWorked
Categorical Columns	Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime, Education (ordinal), JobSatisfaction (ordinal), WorkLifeBalance (ordinal)
Difficulty Level	Medium
Viva Friendliness Score	8.5/10

Why This Dataset Is Suitable

One of the richest column counts of any beginner-safe dataset (35 features), giving strong headroom for multivariate analysis and feature engineering. Directly maps to the HR domain suggested in your guidelines and produces highly explainable business insights for a viva.

Possible Preprocessing Operations

Drop constant columns (EmployeeCount, StandardHours, Over18 are single-valued and should be removed); encode ordinal satisfaction/education scales explicitly rather than treating them as nominal; create a salary-band derived column from MonthlyIncome; create a tenure-vs-role-tenure ratio derived feature.

Possible EDA Analyses

Univariate: attrition rate, income distribution by department. Bivariate: attrition vs overtime, attrition vs job satisfaction. Multivariate: attrition vs (department x overtime); correlation heatmap of all numeric HR metrics; groupby department and job role for average income; pivot table of attrition rate by job role and work-life balance score.

Possible Visualizations

Attrition count bar chart, income distribution histogram by department, boxplot of age by attrition, correlation heatmap, scatter plot of years at company vs monthly income, pie chart of department share, violin plot of job satisfaction by attrition, stacked bar of overtime vs attrition, subplot panel of four key attrition drivers, line chart of average income by years at company.

Possible Insights

Whether overtime is linked to higher attrition; which departments or roles show the highest attrition; how income and promotion gaps relate to attrition; whether work-life balance score predicts attrition; how distance from home affects attrition likelihood.

#3 — Olist Brazilian E-Commerce Public Dataset

Source Link: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Domain	Sales & E-commerce
Number of Records	~100,000 orders (multi-table)
Number of Columns	9 linked CSV files, 40+ combined fields
Numerical Columns	Price, freight value, payment value, payment installments, review score, delivery time (derived), product weight/dimensions
Categorical Columns	Order status, product category, customer state, seller state, payment type
Difficulty Level	Hard
Viva Friendliness Score	7.5/10

Why This Dataset Is Suitable

An exceptionally rich, real commercial dataset that easily clears the 5,000+ record and 15+ column bar once tables are joined. Ideal for showcasing merge/join skills in Pandas, which examiners value highly, but it requires more upfront data engineering than a single-table dataset.

Possible Preprocessing Operations

Merge orders, order_items, payments, customers, and products tables on their key IDs; convert all timestamp columns to datetime; compute a derived delivery-delay column (actual vs. estimated delivery date); handle missing product category translations using the provided translation file; remove duplicate order-item rows.

Possible EDA Analyses

Univariate: order value distribution, review score distribution. Bivariate: delivery delay vs review score, payment type vs order value. Multivariate: review score vs (delivery delay x product category); correlation analysis between freight value, price, and review score; groupby state for average order value; pivot table of monthly order volume by region.

Possible Visualizations

Monthly order volume line chart, top 10 product categories bar chart, review score distribution histogram, boxplot of delivery time by region, correlation heatmap, scatter plot of price vs freight value, pie chart of payment type share, choropleth-style bar of orders by state, subplot grid of seasonal trends, violin plot of review score by delivery delay bucket.

Possible Insights

Whether late deliveries drive down review scores; which regions generate the most revenue; which product categories have the best/worst satisfaction; how payment installments relate to order value; seasonal peaks in order volume.

#4 — World Happiness Report (latest edition)

Source Link: <https://www.kaggle.com/datasets/unsdsn/world-happiness>

Domain	Social / Public Policy
Number of Records	~150 countries/year x multiple years
Number of Columns	10-12
Numerical Columns	Happiness score, GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, perceptions of corruption
Categorical Columns	Country, region/continent

Difficulty Level	Easy
Viva Friendliness Score	8/10

Why This Dataset Is Suitable

Globally recognized, government/UN-backed data that supports clean cross-country comparison and correlation work. Slightly under the 15-column 'preferred' threshold, so derived/engineered columns are needed to maximize depth and marks.

Possible Preprocessing Operations

Combine multiple yearly files into one time-indexed dataframe; standardize country names across years (naming inconsistencies are common); create a derived 'happiness tier' categorical column (Low/Medium/High) by binning the score; handle missing regional labels for a few small countries.

Possible EDA Analyses

Univariate: happiness score distribution. Bivariate: GDP per capita vs happiness score, life expectancy vs happiness score. Multivariate: happiness vs (GDP x social support); correlation matrix across all seven happiness sub-factors; groupby region for average happiness; pivot table of year-over-year score change by region.

Possible Visualizations

World happiness ranking bar chart (top/bottom 10), scatter plot of GDP vs happiness with trendline, correlation heatmap of all factors, boxplot of happiness by region, line chart of global average happiness over years, pie chart of regional share of top-20 countries, violin plot of life expectancy by happiness tier, subplot comparing four sub-factors, radar-style comparison for selected countries.

Possible Insights

Whether GDP alone predicts happiness or social factors matter more; which regions improved or declined most over time; how corruption perception correlates with happiness; outlier countries that are happier or less happy than their GDP would predict; the most influential sub-factor overall.

#5 — Airbnb Listings (NYC, Inside Airbnb / Kaggle mirror)

Source Link: <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>

Domain	Retail / Real Estate Analytics
Number of Records	~49,000
Number of Columns	16

Numerical Columns	Price, minimum nights, number of reviews, reviews per month, availability (days/year), latitude, longitude, calculated host listings count
Categorical Columns	Neighbourhood group, neighbourhood, room type, host name
Difficulty Level	Easy-Medium
Viva Friendliness Score	8/10

Why This Dataset Is Suitable

Comfortably clears both thresholds with a natural mix of geographic, categorical, and numerical data. Pricing and location-based questions translate into very intuitive, presentation-friendly business insights.

Possible Preprocessing Operations

Handle missing 'reviews per month' values (commonly null for zero-review listings; impute as 0, not drop); remove extreme price outliers (e.g., \$0 or unrealistically high listings); convert 'last review' to datetime; create a derived price-per-night-tier categorical column.

Possible EDA Analyses

Univariate: price distribution, room type distribution. Bivariate: price vs room type, price vs neighbourhood group. Multivariate: price vs (room type x neighbourhood group); correlation between price, availability, and review count; groupby neighbourhood for median price; pivot table of average price by room type and borough/area.

Possible Visualizations

Median price by neighbourhood group bar chart, room type share pie chart, price distribution histogram (log-scaled), boxplot of price by room type, correlation heatmap, scatter plot of listings by lat-long colored by price, violin plot of availability by room type, subplot grid of top five neighbourhoods, line chart of review count trend if date-stamped data available.

Possible Insights

Which neighbourhoods command the highest and lowest prices; whether private rooms or entire homes dominate specific areas; how availability relates to pricing strategy; which areas have the most review activity (proxy for demand); outlier hosts with disproportionately large listing counts.

#6 — Healthcare Patient Records (Synthetic Healthcare Dataset)

Source Link: <https://www.kaggle.com/datasets/prasad22/healthcare-dataset>

Domain	Healthcare
Number of Records	10,000+ (depending on version)

Number of Columns	15
Numerical Columns	Age, billing amount, room number, length of stay (derived)
Categorical Columns	Gender, blood type, medical condition, admission type, medication, test results, doctor, hospital, insurance provider
Difficulty Level	Medium
Viva Friendliness Score	7.5/10

Why This Dataset Is Suitable

Meets the 15+ column preference and supports a healthcare-domain project, which the rubric explicitly lists. Synthetic but well-structured, avoiding the privacy and licensing concerns of real patient data.

Possible Preprocessing Operations

Convert admission and discharge dates to datetime and compute a derived length-of-stay column; standardize text-case inconsistencies in categorical fields; bin age into clinically meaningful groups (pediatric, adult, senior); check for and remove duplicate patient-visit records.

Possible EDA Analyses

Univariate: distribution of medical conditions, billing amount distribution. Bivariate: billing amount vs admission type, length of stay vs medical condition. Multivariate: billing vs (insurance provider x admission type); correlation between age, length of stay, and billing; groupby medical condition for average billing; pivot table of admission type by hospital.

Possible Visualizations

Top medical conditions bar chart, billing amount histogram, boxplot of length of stay by condition, correlation heatmap, scatter plot of age vs billing amount, pie chart of insurance provider share, violin plot of billing by admission type, subplot grid of four condition-specific trends, stacked bar of test results by condition.

Possible Insights

Which medical conditions drive the highest billing; whether admission type affects length of stay; which insurance providers are associated with longer or costlier stays; age-group patterns in specific conditions; hospital-level differences in cost or stay duration.

#7 — Global Superstore / Sample Superstore Sales

Source Link: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Domain	Sales & E-commerce / Finance
Number of Records	~9,800-10,000
Number of Columns	18-21
Numerical Columns	Sales, quantity, discount, profit
Categorical Columns	Ship mode, segment, country, region, state, city, category, sub-category
Difficulty Level	Easy
Viva Friendliness Score	8/10

Why This Dataset Is Suitable

A classic, well-documented business analytics dataset purpose-built for dashboards, profitability analysis, and regional comparisons. Slightly less novel than newer datasets but extremely safe and rubric-aligned for Sales/Finance domains.

Possible Preprocessing Operations

Convert order date and ship date to datetime; compute a derived shipping-delay column; create a profit-margin derived column (profit/sales); check for and handle negative-profit outliers; verify no duplicate order-line entries.

Possible EDA Analyses

Univariate: sales distribution, profit distribution. Bivariate: profit vs discount, sales vs category. Multivariate: profit vs (category x region); correlation between sales, discount, and profit; groupby region and category for total sales; pivot table of monthly sales by segment.

Possible Visualizations

Sales by category bar chart, profit by region bar chart, monthly sales trend line chart, scatter plot of discount vs profit, correlation heatmap, boxplot of profit margin by category, pie chart of segment share, violin plot of sales by ship mode, subplot dashboard of four KPIs, stacked bar of sub-category performance.

Possible Insights

Which categories or regions are most/least profitable; whether discounting is eroding margins; seasonal sales patterns; which shipping mode is most commonly used and its cost implication; top and bottom performing sub-categories.

#8 — Students Performance in Exams

Source Link: <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

Domain	Education
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Number of Records	1,000
Number of Columns	8 native (extendable with derived features)
Numerical Columns	Math score, reading score, writing score
Categorical Columns	Gender, race/ethnicity, parental level of education, lunch type, test preparation course
Difficulty Level	Easy
Viva Friendliness Score	7/10

Why This Dataset Is Suitable

Directly matches the Education domain in your guidelines and is extremely approachable, but its native column count (8) is below the 10-minimum requirement, so it qualifies only once you add derived columns. Best suited if you want a lighter, fast-to-clean dataset and are confident engineering 3-4 extra features.

Possible Preprocessing Operations

Create a derived 'average score' column across the three subjects; create a derived pass/fail categorical column using a chosen threshold; create a derived total-score column; encode the five categorical fields; verify there are no missing values (this dataset is typically clean already).

Possible EDA Analyses

Univariate: average score distribution. Bivariate: average score vs test preparation course, average score vs parental education. Multivariate: average score vs (gender x test prep); correlation between the three subject scores; groupby parental education for mean scores; pivot table of pass rate by lunch type and gender.

Possible Visualizations

Average score histogram, boxplot of scores by test preparation course, bar chart of pass rate by gender, correlation heatmap of the three subjects, scatter plot of math vs reading score, pie chart of gender share, violin plot of writing score by parental education, subplot grid comparing all three subjects by group, stacked bar of pass/fail by lunch type.

Possible Insights

Whether completing the test preparation course meaningfully improves scores; how parental education level correlates with performance; whether lunch type (a proxy for socioeconomic status) affects outcomes; which subject shows the widest performance gap across groups; gender-based performance differences.

#9 — Global Air Quality Index by City and Country

Source Link: <https://www.kaggle.com/datasets/rejasupotaro/world-air-quality-index-by-city-and-country>

Domain	Environmental
Number of Records	Varies by version; many exceed 10,000+ city-readings
Number of Columns	10-12
Numerical Columns	AQI value, PM2.5, PM10, CO, NO2, O3 (depending on version)
Categorical Columns	Country, city, AQI category/status
Difficulty Level	Medium
Viva Friendliness Score	7/10

Why This Dataset Is Suitable

Matches the Environmental domain in your guidelines and supports strong time/geography-based analysis, though many public versions sit right at the 10-column minimum, so feature engineering is needed to add analytical depth.

Possible Preprocessing Operations

Standardize country/city name spellings; convert date fields to proper datetime if present; create a derived AQI-severity categorical bucket (Good/Moderate/Unhealthy/Hazardous); handle missing pollutant readings via interpolation or removal; remove duplicate city-date entries.

Possible EDA Analyses

Univariate: AQI distribution. Bivariate: AQI vs country, PM2.5 vs PM10. Multivariate: AQI vs (country x season, if date available); correlation matrix of pollutant measures; groupby country for average AQI; pivot table of AQI severity counts by country.

Possible Visualizations

Top 10 most polluted cities bar chart, AQI distribution histogram, correlation heatmap of pollutants, boxplot of AQI by country, scatter plot of PM2.5 vs PM10, pie chart of AQI severity category share, violin plot of AQI by region, subplot grid of pollutant trends, line chart of AQI over time (if dated).

Possible Insights

Which countries/cities have the most severe air quality; which pollutants are most correlated with high AQI; whether there are regional clusters of poor air quality; seasonal pollution patterns (if time data available); cities that are outliers relative to their country's average.

#10 — IMDB Dataset of 50K Movie Reviews (Sentiment)

Source Link: <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>

Domain	Social Media & Public Sentiment
Number of Records	50,000
Number of Columns	2 native (review text, sentiment label); 10+ only after feature engineering
Numerical Columns	Review length, word count, polarity score (all must be engineered)
Categorical Columns	Sentiment label (positive/negative)
Difficulty Level	Hard
Viva Friendliness Score	6.5/10

Why This Dataset Is Suitable

Excellent record volume and a strong fit for the Sentiment Analysis domain, but its native structure is text-only and requires substantial NLP-style feature engineering (word counts, sentiment scores, length buckets) just to reach the 10-column structured-data minimum. Recommended only if you specifically want a text-analytics angle and are comfortable with that extra workload.

Possible Preprocessing Operations

Engineer numerical features such as review length, word count, average word length, and exclamation/question mark counts; engineer categorical features such as length-bucket and a rule-based or library-based polarity score bucket; clean HTML tags and special characters from raw review text; check class balance between positive and negative labels.

Possible EDA Analyses

Univariate: review length distribution, sentiment class balance. Bivariate: review length vs sentiment, word count vs sentiment. Multivariate: sentiment vs (length bucket x word count bucket); correlation between engineered numeric features; groupby sentiment for average length; pivot table of sentiment by length bucket.

Possible Visualizations

Sentiment class balance bar chart, review length histogram by sentiment, boxplot of word count by sentiment, correlation heatmap of engineered numeric features, word-length distribution violin plot, pie chart of overall sentiment share, scatter plot of length vs word count colored by sentiment, subplot grid of four engineered features, bar chart of most frequent words per class (post text-cleaning).

Possible Insights

Whether longer reviews skew positive or negative; whether punctuation intensity correlates with sentiment; class balance and its implications for analysis; distinguishing vocabulary patterns between positive and negative reviews; whether review length is a usable proxy signal for sentiment.

4. Final Recommendation: Best Dataset for This Project

Recommended Dataset: [Telco Customer Churn \(IBM / WA_Fn-UseC_\)](#)

The Telco Customer Churn dataset is the strongest choice for achieving the highest marks on this specific rubric, for five concrete reasons.

- **It exceeds every hard requirement comfortably.** 7,043 records and 21 columns clear the 500-record and 10-column minimums by a wide margin, and the column count is large enough to support real multivariate analysis without requiring you to manufacture features just to hit a threshold.
- **It has a genuinely balanced mix of data types.** Roughly a third of the columns are numerical (tenure, monthly charges, total charges) and the rest are categorical (16 service, billing, and demographic fields), so groupby operations, pivot tables, and correlation analysis are all natural fits rather than forced exercises.
- **It has a clear, single target variable: Churn.** Every required technique in Task 3, including univariate distribution of churn, bivariate comparisons against contract type or payment method, and multivariate breakdowns combining two or more categorical drivers, maps directly onto a single, easy-to-explain business question: why do customers leave?
- **It supports more than five strong, defensible insights.** Contract type, payment method, tenure, and add-on services (tech support, online security) each independently relate to churn, giving you multiple non-overlapping insights rather than restating the same finding five different ways.
- **It is exceptionally viva-friendly.** Because the dataset models a universally understood business problem (customer retention), you can explain every chart and every insight in plain language during an oral defense without needing to first explain domain jargon, which is a common pitfall with healthcare or environmental datasets.

The one preprocessing step examiners specifically look for is correctly handling the TotalCharges column, which is stored as text and contains a small number of blank values for brand-new customers with zero tenure. Catching and correctly resolving this (rather than blindly force-casting and silently producing errors) is itself a strong, visible demonstration of careful data cleaning for Task 2.

5. Task 1 Write-Up (Ready for Direct Submission)

The section below is formatted to be copied directly into your project notebook as the Task 1 deliverable: Problem Definition & Dataset Selection.

Project Title

Customer Churn Analysis and Retention Insights for a Telecommunications Provider

Problem Statement

Customer churn, the rate at which customers discontinue their subscription, is one of the most significant challenges facing telecommunications companies, where acquiring a new customer is substantially more expensive than retaining an existing one. This project analyzes a real-world telecom customer dataset to identify the demographic, contractual, billing, and service-related factors most strongly associated with churn, with the goal of surfacing patterns that could inform a targeted customer retention strategy.

Project Objectives

1. Perform a complete exploratory data analysis of the Telco Customer Churn dataset, covering data loading, cleaning, and pre-processing.
2. Conduct univariate, bivariate, and multivariate analysis to understand individual feature distributions and their relationships with churn.
3. Apply groupby operations, pivot tables, and correlation analysis to quantify relationships between customer attributes and churn behavior.
4. Produce at least ten meaningful visualizations using Matplotlib and Seaborn to communicate findings clearly.
5. Derive at least five actionable, evidence-based insights that could inform a real customer retention strategy.

Dataset Description

The dataset contains 7,043 customer records and 21 columns describing a fictional telecommunications company's customer base. Each row represents one customer and includes:

- **Demographic attributes:** gender, senior citizen status, partner status, dependents.
- **Account information:** tenure (months with the company), contract type, paperless billing status, payment method.
- **Services subscribed:** phone service, multiple lines, internet service, online security, online backup, device protection, tech support, streaming TV, and streaming movies.
- **Billing information:** monthly charges and total charges.
- **Target variable:** Churn (whether the customer left the company within the last month).

The dataset contains a healthy mix of numerical fields (tenure, monthly charges, total charges) and categorical fields (the remaining 16 service, billing, and demographic columns), satisfying the project's requirement for variety in data types.

Source Information

Dataset Source (Kaggle): <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Originally published by IBM as a sample dataset for customer relationship analytics and widely hosted on Kaggle in CSV format under the filename WA_Fn-UseC_-Telco-Customer-Churn.csv. The dataset is fictional/synthetic, which avoids any personal data or privacy concerns while still reflecting realistic telecom business patterns.

Expected Outcomes

1. A cleaned, fully pre-processed dataset with corrected data types and derived features (such as tenure bands) ready for analysis.
2. A set of at least ten clearly labeled visualizations covering distribution, comparison, and correlation perspectives on churn.
3. A correlation analysis and pivot-table summary identifying which billing and contractual factors most strongly associate with churn.
4. At least five written insights, such as identifying which contract type or payment method carries the highest churn risk, suitable for presentation to a non-technical stakeholder.
5. A short concluding summary translating the analytical findings into potential retention recommendations.

Justification for Dataset Selection

This dataset was selected over alternative domains (healthcare, environmental, education, and social media sentiment) because it uniquely satisfies all four task requirements simultaneously with minimal friction. It exceeds the minimum size requirement (7,043 records against a 500-record minimum) and the minimum feature requirement (21 columns against a 10-column minimum) by a wide margin. Its mix of numerical and categorical data is naturally balanced rather than engineered after the fact, which directly supports every required EDA technique, including groupby analysis, pivot tables, and correlation analysis, without artificial feature creation. Because the dataset centers on a single, intuitive business question, customer retention, every visualization and insight produced from it remains easy to interpret and defend, which is particularly valuable for the documentation and presentation criteria assessed in Task 4.